

Name: _____

Date: _____

FRACTIONS (SECONDARY), (ADDING AND SUBTRACTING FRACTIONS)

LO: I can add and subtract fractions with different denominators by finding a common denominator.

Adding and subtracting fractions

To add or subtract fractions with different **denominators**, first find a **common denominator** (usually the LCM), then add/subtract the numerators.

Steps: 1) Find the LCM of denominators. 2) Convert both fractions. 3) Add/subtract numerators. 4) Simplify.

Key steps

●	$\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$	LCM = 12
●	$\frac{3}{4} - \frac{1}{2} = \frac{3}{4} - \frac{2}{4} = \frac{1}{4}$	LCM = 4
●	Always simplify the final answer	if possible
●	$\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$	cannot add tops and bottoms
●	$\frac{1}{6} + \frac{1}{6} = \frac{1}{6}$	added both top and

Key idea

Same denominator: just add/subtract numerators.
Different: find the LCM first.

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Cross-multiply method (or use LCM)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - same denominator

Calculate $\frac{3}{8} + \frac{2}{8}$

Same denominator: add numerators

$$3 + 2 = 5$$

$$\frac{5}{8}$$

$$= \frac{5}{8}$$

When denominators match, just add the tops.

Example 2 - different denominators

Calculate $\frac{2}{3} + \frac{1}{4}$

LCM of 3 and 4 is 12

$$\frac{2}{3} = \frac{8}{12}; \frac{1}{4} = \frac{3}{12}$$

$$\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$$

$$= \frac{11}{12}$$

Convert both to twelfths, then add.

Example 3 - subtraction

Calculate $\frac{5}{6} - \frac{1}{4}$

LCM of 6 and 4 is 12

$$\frac{5}{6} = \frac{10}{12}; \frac{1}{4} = \frac{3}{12}$$

$$\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$$

$$= \frac{7}{12}$$

Same method: common denominator, then subtract numerators.

Example 4 - mixed numbers

Calculate $1\frac{1}{2} + 2\frac{2}{3}$

Convert: $\frac{3}{2} + \frac{8}{3}$

$$\text{LCM} = 6: \frac{9}{6} + \frac{16}{6} = \frac{25}{6}$$

$$= 4\frac{1}{6}$$

$$= 4\frac{1}{6}$$

Convert to improper fractions, find common denominator, then convert back.

YOUR TURN – PRACTICE (deliberate practice)

1. Same denominator

Add or subtract. Simplify if possible.

- a) $\frac{2}{7} + \frac{3}{7}$
- b) $\frac{5}{8} - \frac{1}{8}$
- c) $\frac{4}{9} + \frac{2}{9}$
- d) $\frac{7}{10} - \frac{3}{10}$

2. Different denominators

Find a common denominator first.

- a) $\frac{1}{2} + \frac{1}{3}$
- b) $\frac{1}{4} + \frac{1}{5}$
- c) $\frac{2}{3} + \frac{1}{6}$
- d) $\frac{3}{4} + \frac{2}{5}$

3. Subtraction with different denominators

Find a common denominator, then subtract.

- a) $\frac{3}{4} - \frac{1}{3}$
- b) $\frac{5}{6} - \frac{2}{3}$
- c) $\frac{7}{8} - \frac{1}{2}$
- d) $\frac{4}{5} - \frac{1}{4}$

4. Mixed numbers

Convert to improper fractions, calculate, convert back.

- a) $1\frac{1}{2} + 2\frac{1}{4}$
- b) $3\frac{1}{3} - 1\frac{1}{2}$
- c) $2\frac{2}{5} + 1\frac{3}{10}$
- d) $4\frac{1}{6} - 2\frac{2}{3}$

5. Mixed addition and subtraction

Calculate. Give answers in simplest form.

- a) $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$
- b) $\frac{5}{6} - \frac{1}{4} + \frac{1}{3}$
- c) $2\frac{3}{4} - 1\frac{1}{3}$
- d) $\frac{7}{10} + \frac{3}{5} - \frac{1}{2}$
- e) $3\frac{1}{2} - \frac{2}{3} - \frac{5}{6}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ ☐

Correct answer: _____

Reason: _____

b) $\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$ ☐

Correct answer: _____

Reason: _____

c) You need a common denominator to add fractions ☐

Correct answer: _____

Reason: _____

d) $\frac{3}{4} - \frac{1}{2} = \frac{2}{2}$ ☐

Correct answer: _____

Reason: _____

e) Always simplify your final answer ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe uses $\frac{2}{3}$ cup of flour and $\frac{3}{4}$ cup of sugar. How much more sugar than flour is used? If you make a triple batch, how much flour and sugar in total?

Expression: _____

Simplified: _____

b) Find three different fractions that add up to 1 where no denominator is repeated and none of the fractions is $\frac{1}{2}$. How many solutions can you find?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

